

Project Milestone 01

CP476 - Internet Computing

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1. Project Title

Course Registration Dashboard

2. Problem Statement

Students often have to plan courses while checking many things at once: prerequisites, course availability, time conflicts, degree requirements, and different course sections. This can make registration stressful and confusing, especially when students are trying to build a full schedule quickly. Administrators also need a simple way to manage courses, sections, terms, and enrollment information.

3. Motivation

The goal is to make course planning and registration easier by putting the main workflow into one dashboard. Instead of forcing users to manually compare course pages, schedules, and requirements, the dashboard would help students search, filter, plan, and register for courses in a more organized way. The project also fits the course because it has a clear front-end workflow, server-side processing, and a relational database backend.

4. Target Users

Primary users: Laurier students who need to search for courses, plan schedules, register, drop, and swap courses.

Secondary users: Administrators who need to add, update, and manage courses, sections, terms, rooms, instructors, and seat availability.

5. App Concept & Scope

The Course Registration Dashboard is a full-stack web application that helps students and administrators manage the course registration process in a more organized way. Students will be able to sign in, search the course catalog, filter courses, view course details, add sections to a plan, check their weekly timetable, identify scheduling conflicts, and submit their selected courses for registration. Administrators will be able to add and manage courses, organize courses by term or semester, view course information, and manage course section details such as instructors, rooms, capacity, and availability.

The application will focus on the main course registration workflow inside the system. It will use its own course, student, section, enrollment, and prerequisite data rather than connecting

directly to Laurier's official registration platform. This keeps the project realistic for the course timeline while still allowing the team to demonstrate a complete front-end interface, server-side processing, and relational database backend.

The main features in scope are student and administrator login, course search and filtering, course detail views, course planning, timetable conflict detection, registration submission, drop and swap actions, and administrator course management. Features outside the initial scope include real Laurier system integration, real tuition or payment handling, official transcript access, and external API features unless approved by the instructor.

7. Feature Priority List

Must Have

- **Authentication Flow:** Automatically prompt the user to log in if they are attempting to access secure areas of the dashboard without an active session.
- **Administrator Tools:** Allow administrators to seamlessly add, edit, and manage courses within the system.
- **Term Organization:** Organize and display courses logically by academic term and semester.
- **Course Details:** Provide a detailed summary view of course information (e.g., description, credits, location, instructor).
- **Core Filtering:** Filter courses by subject and course level (e.g., 100-level, 200-level) to allow basic catalog navigation.

Should Have

- **Calendar Visualization:** View course registrations and potential schedules in a visual, weekly calendar format.
- **Prerequisite Validation:** Filter courses based on the satisfaction of prerequisites (cross-referencing the student's mock academic history).
- **Conflict Resolution:** Filter out or highlight courses based on scheduling conflicts with already carted/enrolled classes.
- **Availability Filtering:** Filter courses to only show those with available, open slots.
- **Smart Error Handling (Co-requisites):** Automatically search for and prompt the user with required co-requisites or linked labs if a registration error occurs at checkout.

Could Have

- **External Insights:** Automatically search and display "Rate My Professor" data when a student views the professor assigned to a course.
- **Customizable UI:** Allow users to set preferences to show and hide specific data columns (e.g., hiding the "Credits" column to save screen space) when browsing the registration table.
- **Export/Print Capabilities:** Allow users to print out a clean, formatted calendar representation of their successfully enrolled courses